Chapter 16. Triple Integrals and Geometry in $\mathbb{R}^3$

Volume, mass, probability, and continuous sums over solid regions

A double integral sums over a two-dimensional region. A triple integral sums over a three-dimensional solid. The change sounds small: replace tiny rectangles by tiny boxes. Conceptually, however, this is a major step. Triple integrals allow us to compute volume, mass, charge, probability, average value, and spatial accumulation over objects in space.

The central geometric skill in this chapter is **describing a solid region by inequalities**. Once the region is described correctly, the integral is often straightforward. Most errors in triple integrals are not integration errors; they are errors in reading the geometry.

This chapter develops triple integrals in rectangular coordinates. Cylindrical and spherical coordinates are treated in the next chapter. For now, the guiding idea is:

$$\boxed{\text{A triple integral is a continuous sum over a solid region.}}$$

Learning goals

After studying this chapter, you should be able to:

1. Interpret $\iiint_E f(x,y,z)\,dV$ as a continuous sum over a solid region E.
2. Compute volume as $\iiint_E 1\,dV$.
3. Set up triple integrals using projections onto coordinate planes.
4. Describe solids with bounds such as $g_1(x,y) \le z \le g_2(x,y)$.
5. Change the order of integration in simple three-dimensional regions.
6. Compute mass, average value, and probability using triple integrals.
7. Use computation to visualize regions and approximate triple integrals numerically.

16.1 From double integrals to triple integrals

Recall the structure of a double integral over a region $R \subset \mathbb{R}^2$:

$$\iint_R f(x,y)\,dA.$$

It is the limit of sums of the form

$$\sum f(x_{ij}^*, y_{ij}^*)\Delta A.$$

For a triple integral over a solid region $E \subset \mathbb{R}^3$, we replace area pieces by volume pieces:

$$\iiint_E f(x,y,z)\,dV = \lim \sum f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)\Delta V.$$

In rectangular coordinates,

$$\Delta V = \Delta x\,\Delta y\,\Delta z, \qquad dV = dx\,dy\,dz.$$

For example,

$$\int_a^b \int_c^d \int_p^q f(x, y, z)\, dz\, dy\, dx$$

means that we integrate first with respect to z, then y, then x.

The guiding picture

A double integral sums tiny columns above an area region. A triple integral sums tiny boxes inside a solid region.

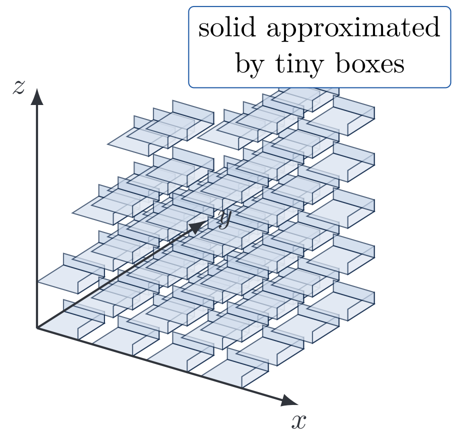

Figure 1: A solid can be approximated by many small rectangular boxes. A triple integral is the limit of such voxel sums.

16.2 Volume as a triple integral

The simplest triple integral occurs when the integrand is 1.

Volume of a solid

If $E \subset \mathbb{R}^3$ is a solid region, then

$$\mathrm{Vol}(E) = \iiint_E 1\, dV.$$

This formula says that volume is the sum of all infinitesimal volume elements dV inside the region. It is the three-dimensional analog of

$$\mathrm{Area}(R) = \iint_R 1\, dA.$$

Example 16.1: A rectangular box

Let

$$E = [0, 2] \times [1, 4] \times [-1, 3].$$

Then

$$\mathrm{Vol}(E) = \int_0^2 \int_1^4 \int_{-1}^3 1\, dz\, dy\, dx.$$

Evaluate from inside to outside:

$$\int_{-1}^3 1\, dz = 4,$$

so

$$\mathrm{Vol}(E) = \int_0^2 \int_1^4 4\, dy\, dx = \int_0^2 12\, dx = 24.$$

Of course, the box has side lengths 2, 3, and 4, so the volume is $2 \cdot 3 \cdot 4 = 24$.

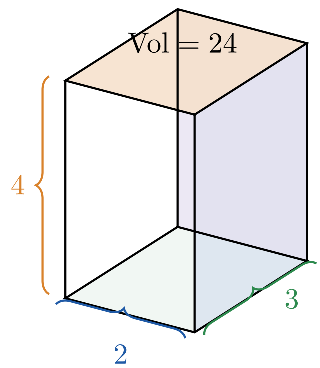

Figure 2: The triple integral of 1 over a rectangular box reproduces length times width times height.

16.3 Solids between two surfaces

A very common solid has the form

$$E = \{(x, y, z) : (x, y) \in D,\ g_1(x, y) \le z \le g_2(x, y)\}.$$

Here D is the projection of E onto the xy-plane. The solid lies above the lower surface $z = g_1(x, y)$ and below the upper surface $z = g_2(x, y)$. Then

$$\iiint_E f(x, y, z)\, dV = \iint_D \left[\int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z)\, dz \right] dA.$$

If D is a Type I region,

$$D = \{(x, y) : a \le x \le b,\ h_1(x) \le y \le h_2(x)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z)\, dz\, dy\, dx.$$

Projection-first method

To set up a triple integral, first choose a projection plane. Then write the outer two bounds for the projection and the inner bounds for the variable perpendicular to that projection.

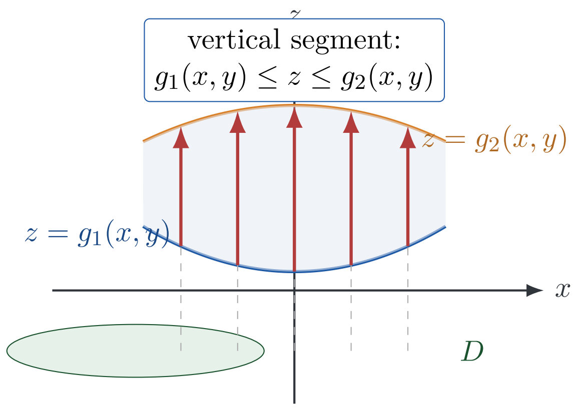

Figure 3: A solid between two surfaces is described by a projection region in the xy-plane and vertical z-bounds.

Example 16.2: Volume under a plane over a triangle

Let E be the tetrahedron in the first octant bounded by the coordinate planes and

$$x + y + z = 1.$$

The region is

$$E = \{(x,y,z) : x \ge 0,\ y \ge 0,\ z \ge 0,\ x + y + z \le 1\}.$$

Project onto the xy-plane. Since z ranges from 0 to $1 - x - y$, the projection is

$$D = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le 1 - x\}.$$

Thus

$$\mathrm{Vol}(E) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\, dz\, dy\, dx.$$

Evaluate:

$$\mathrm{Vol}(E) = \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx = \int_0^1 \frac{(1-x)^2}{2}\, dx = \frac{1}{6}.$$

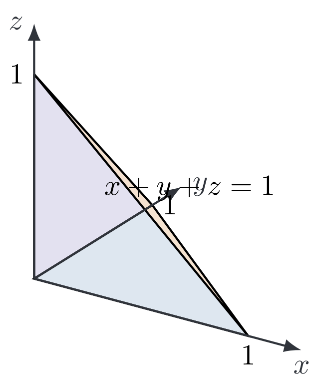

Figure 4: The tetrahedron $x + y + z \le 1$ in the first octant. Its projection onto the xy-plane is a triangle.

16.4 Integrating functions over solids

Volume is only one example. If $f(x, y, z)$ measures a density, intensity, probability density, energy density, or cost density, then

$$\iiint_E f(x, y, z)\, dV$$

sums that quantity over the whole solid.

Example 16.3: A triple integral over the standard tetrahedron

Evaluate

$$\iiint_E y\, dV,$$

where

$$E = \{(x, y, z) : x \geq 0,\ y \geq 0,\ z \geq 0,\ x + y + z \leq 1\}.$$

Using the same bounds as before,

$$\iiint_E y\, dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} y\, dz\, dy\, dx.$$

Since y does not depend on z,

$$\int_0^{1-x-y} y\, dz = y(1 - x - y).$$

Thus

$$\iiint_E y\, dV = \int_0^1 \int_0^{1-x} y(1 - x - y)\, dy\, dx = \int_0^1 \frac{(1-x)^3}{6}\, dx = \frac{1}{24}.$$

Because of symmetry,

$$\iiint_E x\, dV = \iiint_E y\, dV = \iiint_E z\, dV = \frac{1}{24}.$$

16.5 Mass and density in three dimensions

Suppose an object occupies a solid region E and has density $\rho = \rho(x, y, z)$. The mass of a small box is approximately

$$\Delta m \approx \rho(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\Delta V.$$

Summing and passing to a limit gives the mass formula.

Mass from density

If an object occupies $E \subset \mathbb{R}^3$ and has density $\rho(x, y, z)$, then

$$M = \iiint_E \rho(x, y, z)\, dV.$$

Example 16.4: Density increasing with height

Let E be the box

$$0 \le x \le 2, \qquad 0 \le y \le 3, \qquad 0 \le z \le 1,$$

and suppose the density is

$$\rho(x, y, z) = 1 + z.$$

Then

$$M = \int_0^2 \int_0^3 \int_0^1 (1 + z)\, dz\, dy\, dx.$$

First,

$$\int_0^1 (1 + z)\, dz = \frac{3}{2}.$$

Thus

$$M = \int_0^2 \int_0^3 \frac{3}{2}\, dy\, dx = \int_0^2 \frac{9}{2}\, dx = 9.$$

If the density were constant 1, the mass would be the volume 6. Since the density increases with height, the mass is larger.

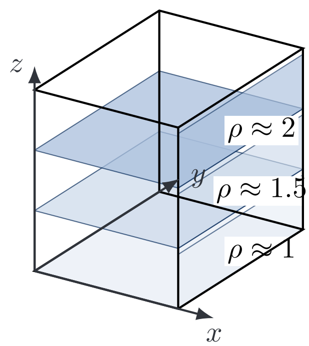

Figure 5: Density $\rho = 1 + z$ is larger in higher horizontal layers, so mass exceeds volume.

16.6 Average value over a solid

For a function $f(x, y, z)$ on a solid E, its average value over E is

$$f_{\text{avg}} = \frac{1}{\text{Vol}(E)} \iiint_E f(x, y, z)\, dV.$$

This is the three-dimensional version of the average value of a function over an interval or a plane region.

Example 16.5: Average height in a tetrahedron

For the standard tetrahedron

$$E = \{(x, y, z) : x, y, z \ge 0,\ x + y + z \le 1\},$$

the average value of z is
$$z_{\text{avg}} = \frac{1}{\text{Vol}(E)} \iiint_E z\,dV.$$
By symmetry, $\iiint_E z\,dV = 1/24$ and $\text{Vol}(E) = 1/6$. Therefore
$$z_{\text{avg}} = \frac{1/24}{1/6} = \frac{1}{4}.$$
The centroid of the tetrahedron is
$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right).$$
This example previews the role of triple integrals in computing centers of mass.

16.7 Other projection directions

Projection onto the xy-plane is common, but not always best. A solid may also be described by projecting onto the yz-plane or the xz-plane.

Projection onto the yz-plane

If
$$E = \{(x, y, z) : (y, z) \in D,\ g_1(y, z) \le x \le g_2(y, z)\},$$
then
$$\iiint_E f(x, y, z)\,dV = \iint_D \left[\int_{g_1(y,z)}^{g_2(y,z)} f(x, y, z)\,dx\right] dA.$$

Projection onto the xz-plane

If
$$E = \{(x, y, z) : (x, z) \in D,\ g_1(x, z) \le y \le g_2(x, z)\},$$
then
$$\iiint_E f(x, y, z)\,dV = \iint_D \left[\int_{g_1(x,z)}^{g_2(x,z)} f(x, y, z)\,dy\right] dA.$$

Do not choose the projection too early

A difficult integral often becomes easier if you project in a different direction. Before writing bounds, sketch the solid and ask: which variable naturally runs between two simple surfaces?

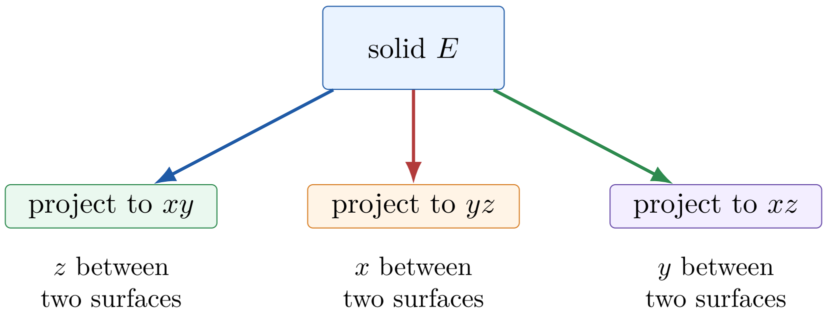

Figure 6: Projection choice determines which variable becomes the innermost integral.

16.8 Changing the order of integration

In double integrals, changing order means rewriting a region in the plane. In triple integrals, changing order means rewriting a solid in space. There are six possible orders:

$$dz\,dy\,dx, \qquad dz\,dx\,dy, \qquad dy\,dz\,dx, \qquad dy\,dx\,dz, \qquad dx\,dz\,dy, \qquad dx\,dy\,dz.$$

The main principle is simple:

$$\boxed{\text{The innermost variable describes a line segment through the solid.}}$$

Example 16.6: The tetrahedron in another order

For the standard tetrahedron

$$E = \{(x, y, z) : x, y, z \geq 0,\ x + y + z \leq 1\},$$

we wrote

$$\iiint_E f(x, y, z)\,dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z)\,dz\,dy\,dx.$$

Now write the order $dx\,dz\,dy$. For fixed y and z,

$$0 \leq x \leq 1 - y - z.$$

The projection onto the yz-plane is

$$0 \leq y \leq 1, \qquad 0 \leq z \leq 1 - y.$$

Therefore

$$\iiint_E f(x, y, z)\,dV = \int_0^1 \int_0^{1-y} \int_0^{1-y-z} f(x, y, z)\,dx\,dz\,dy.$$

Because the tetrahedron is symmetric, all six orders have similar bounds.

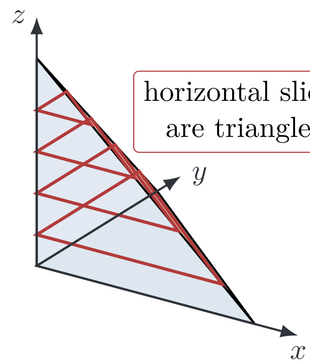

Figure 7: Horizontal slices of the first-octant tetrahedron are triangles. Slicing helps rewrite bounds in a new order.

16.9 Triple integrals and probability

A joint probability density for three continuous random variables X, Y, Z is a nonnegative function $p(x, y, z)$ such that

$$\iiint_{\mathbb{R}^3} p(x, y, z)\, dV = 1.$$

For a solid event $E \subset \mathbb{R}^3$,

$$P((X, Y, Z) \in E) = \iiint_E p(x, y, z)\, dV.$$

This interpretation is essential in statistics and machine learning. Many models use high-dimensional integrals, but the geometric meaning begins here: probability is accumulated density over a region.

Example 16.7: Uniform distribution in a box

Suppose (X, Y, Z) is uniformly distributed in the box

$$0 \le x \le 2, \qquad 0 \le y \le 1, \qquad 0 \le z \le 3.$$

The volume is 6, so the density is $p(x, y, z) = 1/6$ inside the box and 0 outside. The probability of the sub-box

$$0 \le x \le 1, \qquad 0 \le y \le 1, \qquad 1 \le z \le 3$$

is

$$\int_0^1 \int_0^1 \int_1^3 \frac{1}{6}\, dz\, dy\, dx = \frac{2}{6} = \frac{1}{3}.$$

16.10 Numerical triple integration

In practice, many triple integrals cannot be evaluated exactly. Numerical integration is then necessary. For a rectangular box $[a, b] \times [c, d] \times [p, q]$, a midpoint rule approximation is

$$\iiint_E f(x, y, z)\, dV \approx \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=1}^{\ell} f(\bar{x}_i, \bar{y}_j, \bar{z}_k) \Delta x \Delta y \Delta z.$$

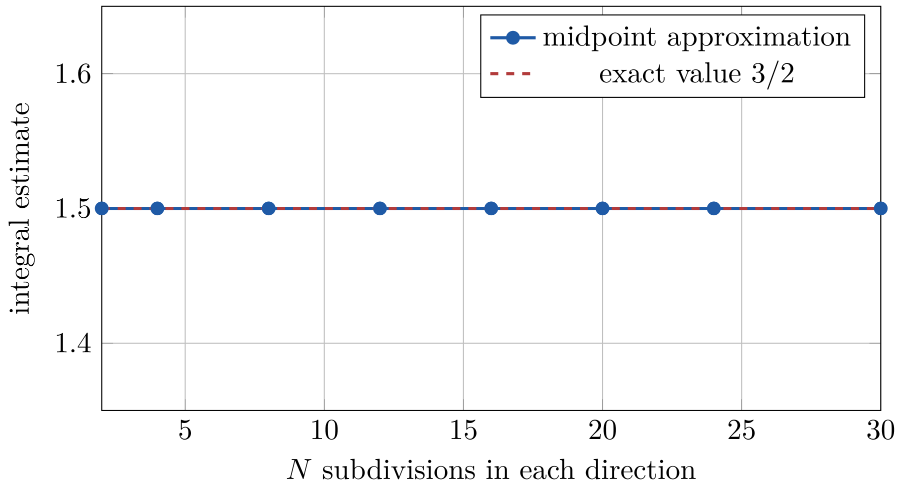

Figure 8: For $f(x, y, z) = x + y + z$ over the unit cube, the midpoint rule is exact for every grid size because the function is linear.

For nonlinear functions, the approximations generally improve as the grid is refined.

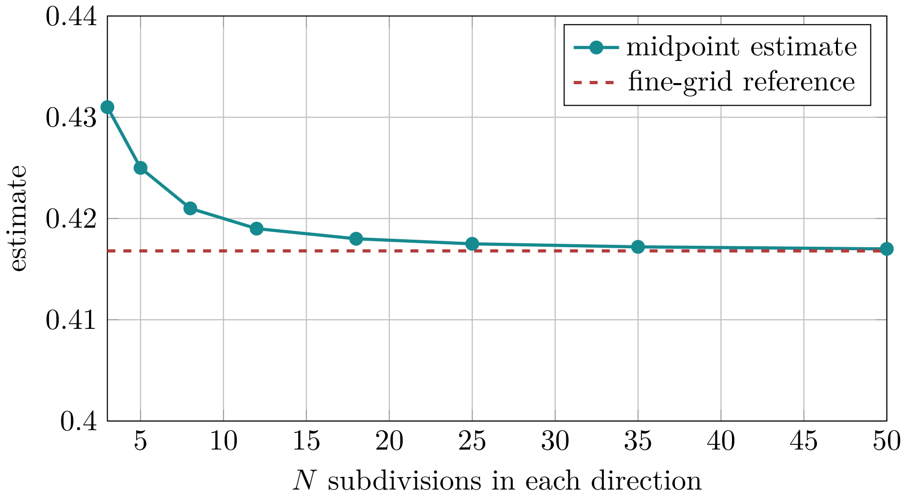

Figure 9: Midpoint approximations for a nonlinear function over the unit cube stabilize as the grid is refined.

16.11 Monte Carlo integration in three dimensions

Monte Carlo integration estimates an integral by random sampling. Suppose E is a box of volume V and we want

$$I = \iiint_E f(x, y, z)\, dV.$$

If $P_1, \ldots, P_N$ are uniformly sampled from E, then

$$I \approx V \cdot \frac{1}{N} \sum_{i=1}^{N} f(P_i).$$

Monte Carlo methods become especially important in high-dimensional statistics, Bayesian inference, uncertainty quantification, and machine learning.

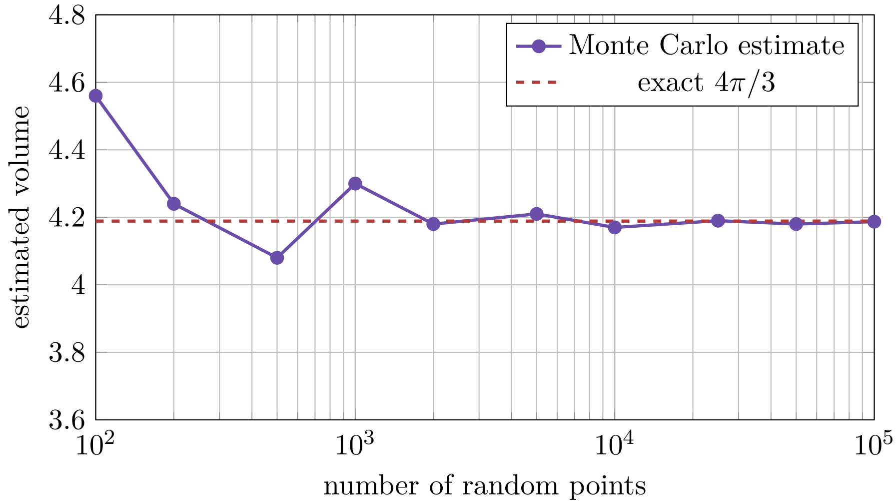

Figure 10: Monte Carlo estimation of the volume of the unit ball. Random estimates fluctuate but tend to stabilize with many samples.

16.12 Moments and centers of mass

Triple integrals also compute balance points. Suppose an object occupies E with density $\rho(x, y, z)$ and mass

$$M = \iiint_E \rho(x, y, z)\, dV.$$

The moments about the coordinate planes are

$$M_{yz} = \iiint_E x\rho(x, y, z)\, dV, \qquad M_{xz} = \iiint_E y\rho(x, y, z)\, dV, \qquad M_{xy} = \iiint_E z\rho(x, y, z)\, dV.$$

The center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_{yz}}{M}, \frac{M_{xz}}{M}, \frac{M_{xy}}{M}\right).$$

If ρ is constant, the center of mass is the centroid of the region.

Example 16.8: Centroid of the standard tetrahedron

For the standard tetrahedron, by symmetry and the computation above,

$$\bar{x} = \bar{y} = \bar{z} = \frac{1}{4}.$$

Thus the centroid is

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right).$$

This point is closer to the origin than to the opposite face because the tetrahedron is widest near the coordinate planes.

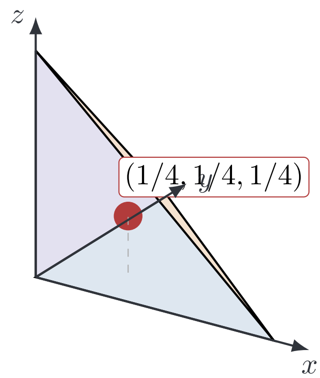

Figure 11: The centroid of the first-octant tetrahedron is $(1/4, 1/4, 1/4)$.

16.13 Triple integrals in the age of data and AI

Triple integrals appear naturally in modern applied mathematics, statistics, and machine learning.

Spatial data and voxel models

Medical imaging, climate models, fluid simulations, and three-dimensional computer vision often represent space using voxels. A voxel is a small three-dimensional box. A triple integral is the mathematical limit of summing quantities over voxels.

Examples include total mass in a scanned object, total heat in a region, average intensity in a three-dimensional image, and probability that a random spatial point lies in a target region.

Probability and Bayesian inference

For three parameters $\theta_1, \theta_2, \theta_3$, a posterior density $p(\theta_1, \theta_2, \theta_3 \mid \text{data})$ satisfies

$$\iiint p(\theta_1, \theta_2, \theta_3 \mid \text{data})\, d\theta_1\, d\theta_2\, d\theta_3 = 1.$$

A probability statement about the parameters is a triple integral over a region in parameter space.

Continuous loss and risk

In statistical learning, expected loss may be written as an integral over feature space. In three variables, a risk functional may look like

$$R(w) = \iiint L(w; x, y, z) p(x, y, z)\, dV.$$

This is a triple integral of loss weighted by a probability density.

16.14 Common mistakes

Mistake 1: Forgetting the projection

Bounds such as $0 \le z \le g(x, y)$ are incomplete unless the allowed (x, y) values are described.

Mistake 2: Using the wrong upper surface

Always check which surface is above on the projection region.

Mistake 3: Mixing the order of bounds and differentials

In $dz\,dy\,dx$, the z-bounds come first, then y-bounds, then x-bounds.

Mistake 4: Treating dependent variables as constants incorrectly

When integrating with respect to z, expressions involving x and y are constant, but the z-bounds may depend on x and y.

Mistake 5: Assuming every solid is a box

Most real regions require variable bounds.

Mistake 6: Not checking units

If f has units of density, then $f\,dV$ has units of mass.

16.15 Summary

A triple integral is a continuous sum over a solid region:

$$\iiint_E f(x, y, z)\,dV.$$

The most important special cases are:

$$\mathrm{Vol}(E) = \iiint_E 1\,dV, \qquad M = \iiint_E \rho(x, y, z)\,dV, \qquad f_{\text{avg}} = \frac{1}{\mathrm{Vol}(E)} \iiint_E f(x, y, z)\,dV.$$

The most common rectangular-coordinate setup is

$$E = \{(x, y, z) : (x, y) \in D,\ g_1(x, y) \le z \le g_2(x, y)\},$$

which gives

$$\iiint_E f(x, y, z)\,dV = \iint_D \left[\int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z)\,dz \right] dA.$$

The key skill is not merely integration. The key skill is **turning geometry into correct bounds**.

Exercises

Conceptual exercises

1. Explain why $\iiint_E 1\, dV$ gives the volume of E.
2. What is the projection of a solid onto the xy-plane?
3. In the integral
$$\int_0^2 \int_0^{4-x} \int_{x+y}^{5} f(x,y,z)\, dz\, dy\, dx,$$
describe the solid region in words.
4. Why might it be easier to project a region onto the yz-plane instead of the xy-plane?
5. Explain the difference between mass and volume in terms of triple integrals.
6. What does it mean for $p(x,y,z)$ to be a joint probability density?

Computational exercises

7. Evaluate $\int_0^1 \int_0^2 \int_0^3 (x+y+z)\, dz\, dy\, dx$.
8. Find the volume of the box $-1 \le x \le 2$, $0 \le y \le 4$, $1 \le z \le 5$.
9. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\, dz\, dy\, dx$.
10. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} x\, dz\, dy\, dx$.
11. Let E be the solid under $z = 4 - x^2 - y^2$ and above the rectangle $0 \le x \le 1$, $0 \le y \le 1$. Set up a triple integral for the volume of E.
12. Let E be the solid between $z = x^2 + y^2$ and $z = 4$. Set up the volume integral in rectangular coordinates using projection onto the xy-plane. Do not evaluate.
13. Let E be the tetrahedron $x, y, z \ge 0$, $x + y + z \le 2$. Find its volume.
14. Let $E = [0,1]^3$ and $\rho(x,y,z) = 1 + x + y + z$. Find the mass of E.
15. Find the average value of $f(x,y,z) = z$ over the unit cube $[0,1]^3$.

Geometry and setup exercises

16. Describe the solid represented by
$$0 \le x \le 1, \qquad 0 \le y \le 1 - x, \qquad 0 \le z \le 2 - x - y.$$
17. Rewrite the tetrahedron integral
$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x,y,z)\, dz\, dy\, dx$$
in the order $dy\, dz\, dx$.

18. Set up, but do not evaluate, an integral for the volume of the solid above $z = 0$, below $z = 9 - x^2 - y^2$, and above the first quadrant of the xy-plane.
19. Set up a triple integral for the mass of the solid in Exercise 18 if $\rho(x, y, z) = z + 1$.
20. Let E be the region bounded by $z = 0$, $z = 1 - y$, $x = 0$, $x = 2$, and $y = 0$. Write $\iiint_E f(x, y, z)\, dV$ as an iterated integral.

Computational lab exercises

21. Use a midpoint rule on a grid to approximate
$$\int_0^1 \int_0^1 \int_0^1 e^{-(x^2+y^2+z^2)}\, dV.$$
22. Use random sampling to estimate the volume of the region $x^2 + y^2 + z^2 \leq 1$.
23. Use random sampling to estimate the volume of the tetrahedron $x, y, z \geq 0$, $x + y + z \leq 1$.
24. Plot horizontal slices of the solid under $z = 4 - x^2 - y^2$ and above the xy-plane.
25. Write a function that estimates $\iiint_E f\, dV$ over a box using the midpoint rule.

Python lab

A companion Python lab for this chapter should include:

- plotting basic solid regions,
- voxel approximations of volume,
- midpoint approximation for triple integrals,
- Monte Carlo estimation of volumes,
- probability integrals over 3D regions,
- center-of-mass computations for simple solids.